

Predictive Modeling of Residential Energy Consumption in Los Angeles Using Integrated Housing, Geospatial, and Socioeconomic Data

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Abstract

This project aims to analyze household energy consumption patterns across Los Angeles by integrating and modeling datasets collected. I combined three key sources: UCLA Energy Atlas data, Craigslist housing listings (scraped for house listings), and google's geocoding API. Using CatBoost regression models trained on UCLA datasets, I predicted three features for each house: median energy usage (kWh), per capita usage (kWh), and estimated income range. These predictions I compiled into a unified dataset (houses_final.csv) enriched with location-specific solar potential values.

Our analysis reveals several important insights. Energy consumption per capita increases with estimated income. Home price (can be rent) is a statistically significant predictor of energy usage per capita. A final multivariate linear regression shows structural features such as square footage, number of bathrooms, and bedrooms are the strongest drivers of energy consumption and when controlling the size of the house, price starts to decrease median the energy consumption, while solar potential is negatively correlated with usage. These findings can be helpful for targeting energy efficiency and policy planning.

Motivation

Energy plays a critical role in modern life, powering everything from heating and transportation to advanced technologies. With increased time spent at home, particularly post-COVID, residential energy consumption has grown significantly, making it essential to understand the factors influencing household energy use. This analysis explores how geospatial location, house size, property value, and solar potential affect energy consumption (positively, negatively, or not at all), aiming to identify patterns and insights to support effective energy efficiency and sustainability policies.

Data Sources

Initially, I tried to obtain household-level energy consumption data. However, due to privacy restrictions, accessing such granular data was not possible. Thus, I shifted my focus to using aggregated data. This led me to explore publicly available datasets that provide aggregated energy consumption data. After searching the internet, I discovered the UCLA Energy Atlas.

Dataset 1 - UCLA Energy Atlas

<https://energyatlas.ucla.edu/>

The UCLA Energy Atlas provides five datasets detailing household attributes related to energy consumption across specific geographic areas in Los Angeles, including income groups, building type, age, size, solar adoption, and annual energy usage (in kWh). Together, they form a robust dataset of 38,962 rows, pretty enough for this analysis.

The datasets and their respective columns are as follows:

- **usage_income_kwh.csv:** geo_id, year, income_quintile, income_range, usage, pop, usetype, usage_percap, name
- **usage_bld_kwh.csv:** geo_id, sqft, usage, usage_med, usage_med_sqft, usetype, year, name, solar_potential, pop, usage_percap
- **usage_vintage_kwh.csv:** geo_id, sqft, usage, usage_med, usage_med_sqft, usetype, vintage_id, year, name
- **usage_sqft_kwh.csv:** geo_id, sqft, sqft_id, usage, usage_med, usage_med_sqft, usetype, year, name
- **geos.csv:** geo_id, name, pop, pop_sqmi, med_income, pct_own, pct_rent, year, enviroscreen_tracts, sqmi, solar_pot, net_solar_pot, count

Sample output from **usage_income_kwh** dataset:

geo_id	year	income_quin	income_range	usage	pop	usetype	usage_percap	name
cities_1	2016	3	\$56,290-\$74,167	6173795	3224	res	1914.9488	agoura hills
cities_1	2016	4	\$74,167-\$99,028	17494261	6323	res	2766.7659	agoura hills
cities_1	2016	5	> \$99,028	46413877.1	14504	res	3200.0743	agoura hills

As you can see, one record shows that households in the **Agoura Hills** city with an average income **above \$99,028** consume around **46,413,766.1 kWh annually**, or **a median annual residential consumption of 3,200.07 kWh per capita**.

The **geo_id/name** fields vary across observations. To enable accurate household-level matching, I used the Google Geocoding API to extract and standardize geographic labels such as neighborhood, ZIP code, county, and city. The main goal is to support more granular spatial analysis and enhance the aggregation of energy usage data.

Dataset 2 - Google Maps Platform - Geocoding API

<https://developers.google.com/maps/documentation/geocoding/overview>

This API allows us to input partial place names, such as “Agoura Hills,” and return detailed geographic data such as postal code, neighborhood, city, and county through forward geocoding.

Sample output of using the API with ‘api_locations.csv’ dataset:

geo_id	name	query_for_api	postal_code	city	county	neighborhood	lat	lng
cities_1	agoura hills	agoura hills, CA	91301	Agoura Hills	Los Angeles County		34.1533395	-118.7616764
cities_10	carson	carson, CA		Carson	Los Angeles County		33.8316745	-118.281693
cities_100	camarillo	camarillo, CA		Camarillo	Ventura County		34.2163937	-119.0376023

Dataset 3 - Scraping Craigslist

<https://losangeles.craigslist.org/search/hhh#search=2~gallery~0>

I will scrape Craigslist to collect housing listings queried by the 150 most frequently occurring ZIP codes, neighborhoods, and cities in Los Angeles.

First few rows from the **list of most frequently occurring geographic names**:

	name	Count	Zip_Code
0	agoura hills	162	91301
1	pico rivera	162	None
2	azusa	162	None
3	hawaiian gardens	162	None

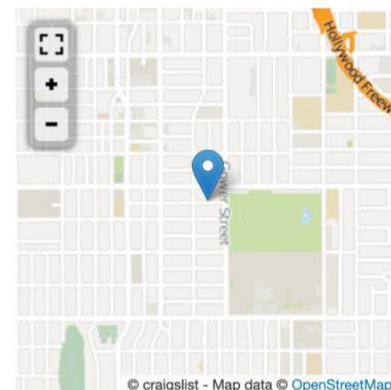
An example screenshot of a craigslist listing that is being scraped

\$1,795 / 1br - 650ft² - Eleanor Apts...Gorgeous Newly Remodeled 1 Bedroom...Amazing Location! (Hollywood)

6046 Eleanor Ave., Los Angeles, CA 90038



Eleanor Apartments has a one bedroom available. Don't miss out on this great unit, located minutes away from the Wilshire Country Club, 101 freeway and the Los Angeles Community



6046 Eleanor Ave.

[google map](#)

1BR / 1Ba 650ft²

application fee details: \$50

Code – Technical Solution

JN 1 - UCLA_data_processing_and_API.ipynb

Important: To run all of the code successfully, only five files are required. These datasets are available from UCLA's Energy Atlas and can be downloaded from:

<https://energyatlas.ucla.edu/data>

The required files are: **geos.csv**, **usage_bld_kwh.csv**, **usage_income_kwh.csv**, **usage_sqft_kwh.csv**, **usage_vintage_kwh.csv**

This is the first JN to start the process. In this notebook, I processed five UCLA energy datasets mentioned, merging them, creating a sparse dataset named **'ucla_merged.csv'**.

Sample output of 'ucla_merged.csv':

geo_id	sqft	usage	usage_med	usage_med_sqft	usetype	year	name	solar_potential	pop	usage_percap	source	vintage_id	income Quintile	income_range	sqft_id	pop_sqmi	med_income	pct_own	pct_rent	enviroscreen_tracts	sqmi	solar_pot	net_solar_pot	count
0	cities_1	1253.0	masked	masked	masked	agriculture	2016	agoura hills	NA	NA	masked	usage_bld_kwh	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
1	cities_1	4802115.0	65825608.0000	121901	9.7676	commercial	2016	agoura hills	NA	NA	NA	usage_bld_kwh	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
2	cities_1	8906397.0	43707916.6000	9136	3.9839	condo	2016	agoura hills	NA	NA	NA	usage_bld_kwh	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
3	cities_1	734698.0	masked	masked	masked	industrial	2016	agoura hills	NA	NA	masked	usage_bld_kwh	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
4	cities_1	100192.0	masked	masked	masked	institutional	2016	agoura hills	NA	NA	masked	usage_bld_kwh	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
5	cities_1	742420.0	2010646.5000	70823	3.2102	multi_family	2016	agoura hills	NA	NA	NA	usage_bld_kwh	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
6	cities_1	4199.0	masked	masked	masked	other	2016	agoura hills	NA	NA	masked	usage_bld_kwh	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
7	cities_1	46700.0	masked	masked	masked	residential_other	2016	agoura hills	NA	NA	masked	usage_bld_kwh	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
8	cities_1	0.0	masked	masked	masked	residential_uncat	2016	agoura hills	NA	NA	masked	usage_bld_kwh	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
9	cities_1	14557697.0	70081933.1000	9369	4.0788	res_total	2016	agoura hills	NA	20236.0	3463.2305	usage_bld_kwh	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

Following that the code cleans the datasets to create a unified view of residential energy consumption and get the location data. Then, the code creates the queries to be used in the API with the location information, creating **'api_locations.csv'**.

Using the Geocoding API

To use the Google Geocoding API, you'll need to generate your own API key. For security reasons, my API key is not included in the notebook. You can follow the official instructions provided by Google to obtain your own key:

<https://developers.google.com/maps/documentation/geocoding/start>

Once you have your key, simply put your API key in the 11th code block, assign it to the object **'my_api_key'**.

Then just run the codeAPI to generate the **ucla_final.csv** file which has the extra geographic data generated with the geocoding API.

Sample output of 'ucla_final.csv':

ucla_final

geo_id	sqft	usage	usage_med	usage_med_sqft	usetype	year	name	solar_potential	pop	usage_percap	source	vintage_id	income Quintile	income_range	sqft_id	pop_sqmi	med_income	pct_own	pct_rent	enviroscreen_tracts	sqmi	solar_pot	net_solar_pot	count	query	postal_code	city	county	neighborhood	lat	lng	
cities_1	1253.0	masked	masked	masked	agriculture	2016	agoura hills			masked	usage_bld_kwh															agoura hills, CA	91301	Agoura Hills	Los Angeles County		34.1533395	-118.7616764
cities_1	4802115.0	65825608.0000	121901	9.7676	commercial	2016	agoura hills				usage_bld_kwh															agoura hills, CA	91301	Agoura Hills	Los Angeles County		34.1533395	-118.7616764
cities_1	8906397.0	43707916.6000	9136	3.9839	condo	2016	agoura hills				usage_bld_kwh															agoura hills, CA	91301	Agoura Hills	Los Angeles County		34.1533395	-118.7616764
cities_1	734698.0	masked	masked	masked	industrial	2016	agoura hills			masked	usage_bld_kwh															agoura hills, CA	91301	Agoura Hills	Los Angeles County		34.1533395	-118.7616764

The final task is to collect individual house data, which I will do by scraping Craigslist.

JN 2 - Craigslist_Scraping.ipynb

This is the second JN, please make sure you have run all code blocks and successfully created the `ucla_final.csv` in the first JN before starting this. First, the code reads the `'ucla_final.csv'` and extracts the top 150 locations from it. Defining a function named `scrape_craigslist` to search Craigslist using a combination of location name and ZIP code (if present). Then, iterating through the top 150 locations to collect listings' **titles** and **URLs** using the function. Save the outcome in a file named `'craigslist_listings.csv'`.

Sample Output of the `'craigslist_listings.csv'` dataset:

	title	link	search_name	search_zip
0	Gorgeous 1 bed / 1 bath! 763 Sq Ft! Must see!	https://losangeles.craigslist.org/sfv/apa/d/agoura-hills-gorgeous-bed-bath-763-sq/7842332144.html	agoura hills	91301
1	Beautiful grounds are waiting for you at The Lexington Apartment Homes	https://losangeles.craigslist.org/sfv/apa/d/agoura-hills-beautiful-grounds-are/7841622716.html	agoura hills	91301
2	Bask in luxury at The Lexington Apartment Homes. The finest in upscale	https://losangeles.craigslist.org/sfv/apa/d/agoura-hills-bask-in-luxury-at-the/7841879707.html	agoura hills	91301
3	Your haven from a busy lifestyle! Calm and quiet 1 bed / 1 bath	https://losangeles.craigslist.org/sfv/apa/d/agoura-hills-your-haven-from-busy/7841356523.html	agoura hills	91301

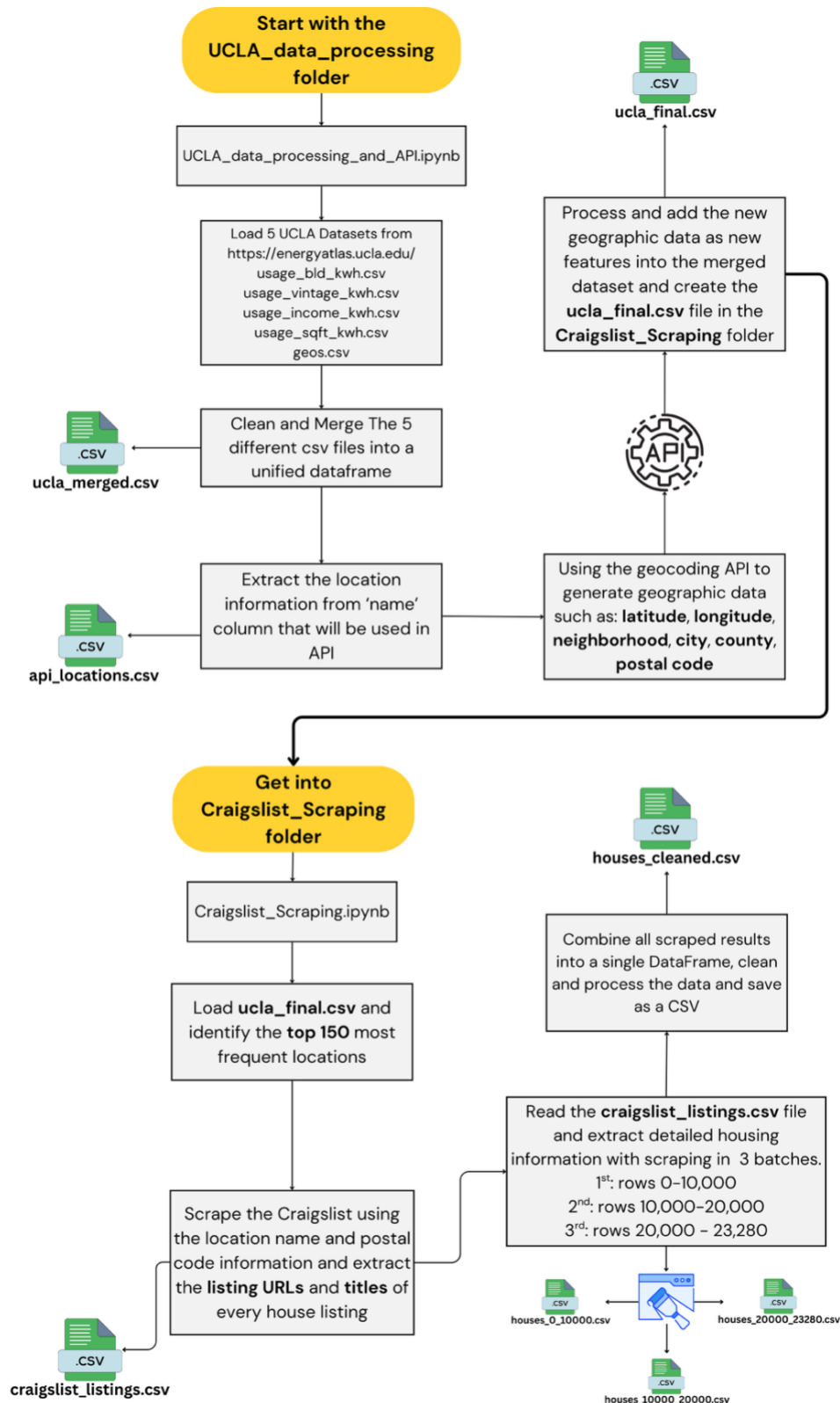
Then, using the URLs from the `'craigslist_listings.csv'`, the code queries individual Craigslist listing URLs and parses the HTML content of that URL to extract structured data. To collect housing data from Craigslist, the scraping process was executed in three separate batches based on row ranges: **0–10,000**, **10,000–20,000**, and **20,000–23,280**, and save them as `houses_new_‘batch_range’`, so first batch is saved as `houses_new_0_10000.csv`. Each batch was processed in a separate run using the `scrape_craigslist_listing` function, with a total runtime of approximately **8 hours** for all batches. For politeness, it includes a **delay of 1.2 seconds** between each request. The scraping function extracts structured information such as **rent price**, **title**, **location**, **number of bedrooms** and **bathrooms**, **square footage**, and **listing description** from individual Craigslist listing pages, using BeautifulSoup.

After scraping, the three batch files were merged into a single dataset. The script then removes exact duplicates by checking for repeated combinations of address and price, ensuring each house appears only once. Listings with excessive missing data, specifically those missing more than two key attributes such as square footage, number of bedrooms, bathrooms, or location details are also filtered out. The resulting cleaned dataset is saved as `houses_cleaned.csv`, which serves as the foundation for the energy usage prediction analysis.

Sample output from `houses_new_10000_20000.csv` dataset:

title	price	bedrooms	bathrooms	square_foot	address	latitude	longitude	description	search_name	search_zip	url
Experience the Ultimate in Comfort: 2 BR with Private Patio (Carson)	3,500	2	2	1172	615 E Carson	33.831675	-118.26553	QR Code	carson		https://losangeles.craigslist.org/sfv/apa/d/experience-the-ultimate-in-comfort-2-br-with-private-patio-carson/7841622716.html
The pinnacle of comfort: Exceptional 3 BR spaces available. (Carson)	4,199	3	2	1325	285 E Del Am	33.846401	-118.27501	QR Code	carson		https://losangeles.craigslist.org/sfv/apa/d/the-pinnacle-of-comfort-exceptional-3-br-spaces-available-carson/7841879707.html
Purrfect home for you & your pet! 1 BR / 1 BA in Carson. (Carson)	3,253	1	1	931	285 E Del Am	33.846401	-118.27501	QR Code	carson		https://losangeles.craigslist.org/sfv/apa/d/purrfect-home-for-you-your-pet-1-br-1-ba-in-carson-carson/7841356523.html
Downtown Carson (Carson)	949				215th PL, Cai	33.833217	-118.2405	QR Code	carson		https://losangeles.craigslist.org/sfv/apa/d/downtown-carson-carson/7841622716.html

FLOWCHART FOR DATASET GENERATION – JN1 & JN2:



JN 3 - Predicting_energy.ipynb

Data Preprocessing

Don't start this without running the other JNs first. Get into the **Predicting_Energy** folder. I begin by loading the combined **ucla_final.csv** dataset. To standardize the data. Placeholder "masked" entries are replaced with pandas NaN. Numeric columns such as **usage_med**, **usage_percap**, **sqft**, and **usage** are explicitly converted to numeric values using **pd.to_numeric**. Categorical fields (**city**, **county**, **postal_code**, etc.) are filled with 'Unknown' if missing to prepare the data for ML training.

Predictive Modeling Using CatBoost (Gradient Boosting Decision Trees)

To predict energy consumption and environmental attributes for houses, I trained five distinct CatBoostRegressor models, each aligned with a specific dataset source:

I trained CatBoost regression models on four datasets, with **usage_sqft_kwh** (R-square=0.76) and **usage_vintage_kwh** (R-square=0.21) predicting median usage (**usage_med**) via weighted averaging. **Usage_income_kwh** (R-square=0.23) predicted per capita usage (**usage_percap**), estimating household income from rent prices. The geos dataset required no modeling; solar potential was directly merged into the dataset, treating negative values as missing due to invalid interpretations. **Usage_bld_kwh** was excluded because of poor performance (negative R-square). I created **houses_final.csv** by enriching housing listings from **houses_cleaned.csv** with three new features. **usage_med_predicted** was computed using a weighted average of two CatBoost models trained on **usage_sqft_kwh** and **usage_vintage_kwh**. **usage_percap_predicted** was added using a third CatBoost model trained on **usage_income_kwh**, with income range inferred from predicted rent. Lastly, **solar_pot** values were merged from the geos dataset using location name matching.

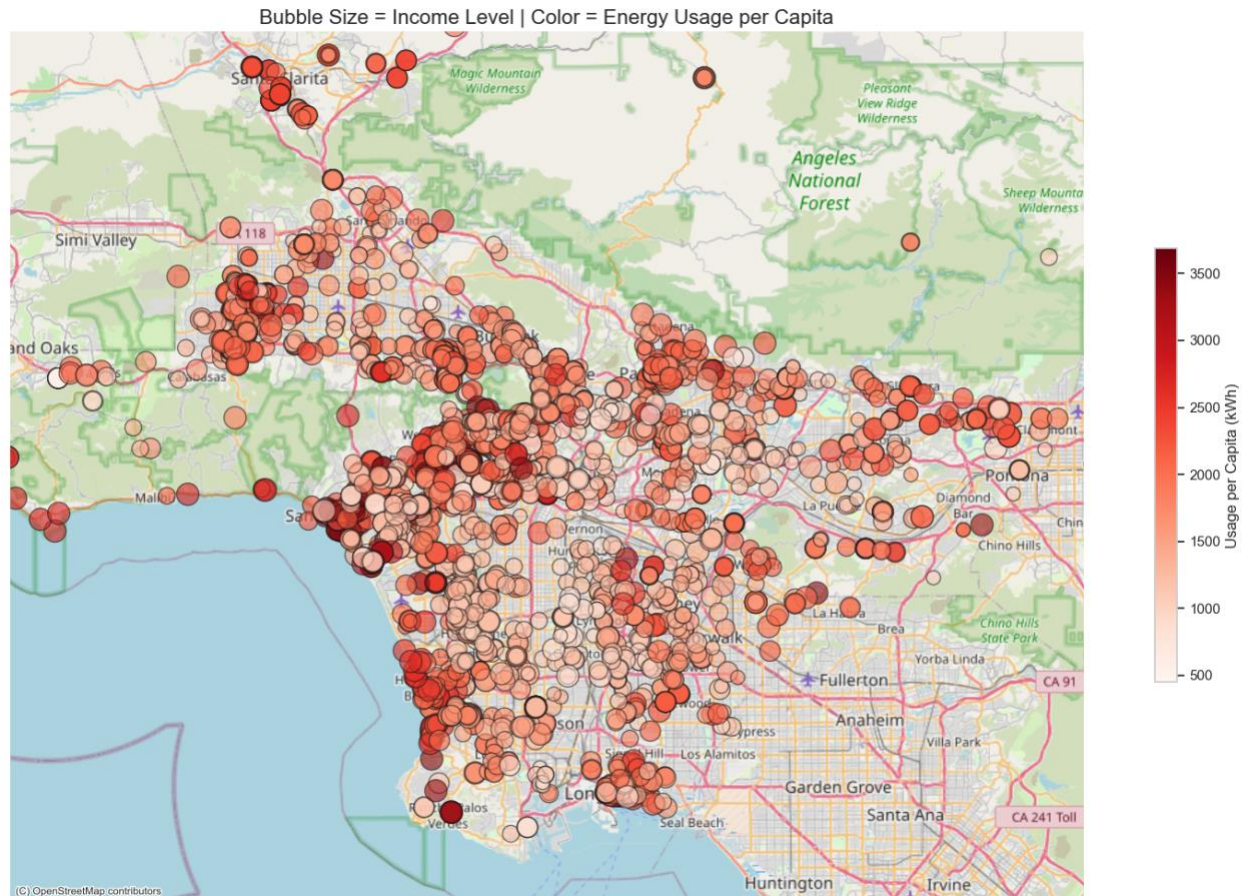
Workflow of Predictions:



Analysis

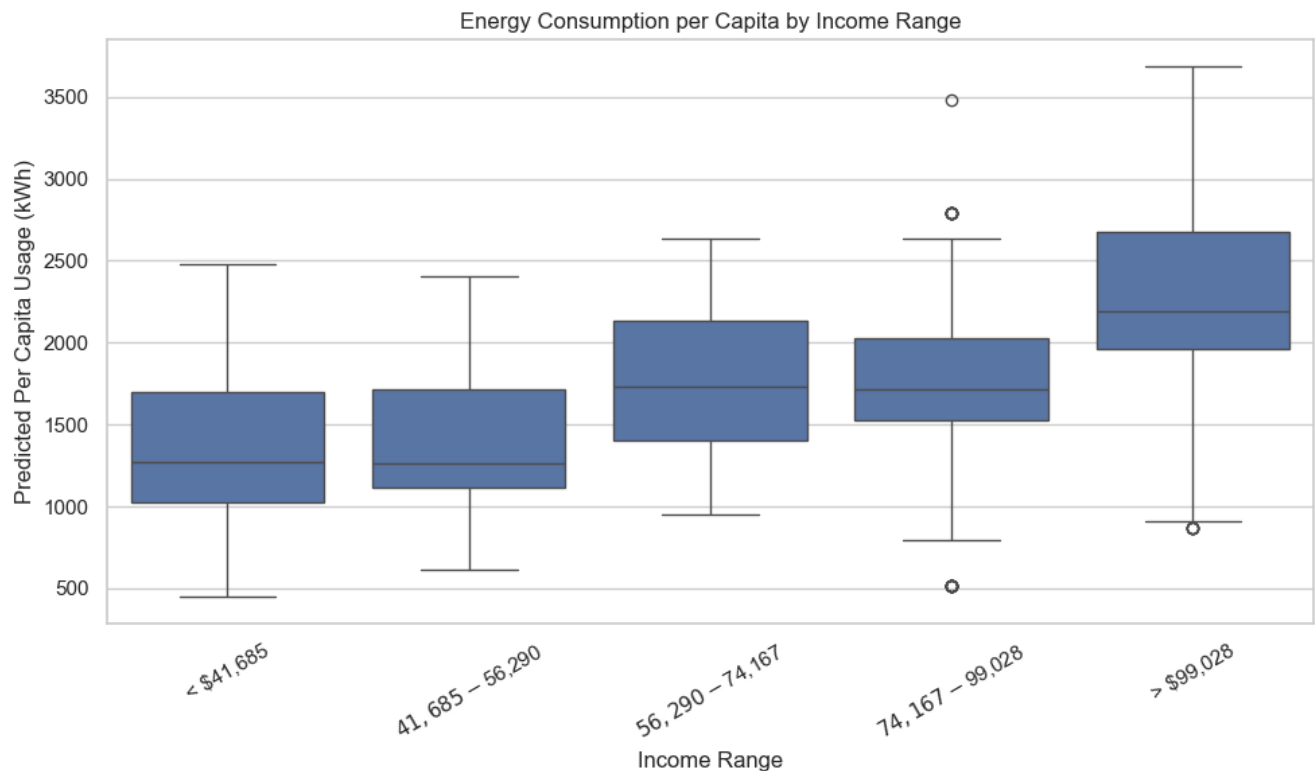
To start my analysis, I first created a bubble map using the latitude and longitude of each house. The color intensity of each bubble represents the predicted per capita energy usage, darker color indicates higher consumption, and bubble size reflects the estimated income level.

Bubble Map with Energy Usage Per Capital and Income Level



It seems like income range and energy usage per capita seems to have a correlation. Especially it looks like affluent neighborhoods and cities seems to have darker bubbles which means a positive correlation with income and energy usage. Therefore, I will analyze how predicted per capita energy usage varies by income range. After filtering out missing data, the code uses a box plot to show the distribution of usage across income brackets, helping reveal whether energy consumption patterns differ by income levels.

Income Range and Usage Per Capita



As you can see, the analysis reveals a positive correlation between income and per capita energy consumption. Notably, there are two significant breaking points, energy usage rises sharply when income surpasses \$56,290 and again beyond \$99,028, suggesting higher-income households tend to consume more energy.

Under "Price and Usage Per Capita - Regression," I built a linear regression model to test if house price predicts per capita energy usage. Price significantly explains 18% of usage variance, with each unit price increase linked to a 0.2462 kWh rise in energy consumption per capita.

Result of the OLS Linear Regression model

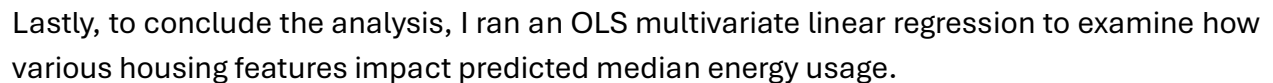
OLS Regression Results

Dep. Variable:	usage_percap_predicted	R-squared:	0.184
Model:	OLS	Adj. R-squared:	0.184
Method:	Least Squares	F-statistic:	786.9
Date:	Mon, 05 May 2025	Prob (F-statistic):	2.77e-156
Time:	20:56:43	Log-Likelihood:	-26705.
No. Observations:	3483	AIC:	5.341e+04
Df Residuals:	3481	BIC:	5.343e+04
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1142.3399	23.741	48.117	0.000	1095.792	1188.887
price	0.2462	0.009	28.052	0.000	0.229	0.263

Omnibus:	61.958	Durbin-Watson:	0.636
Prob(Omnibus):	0.000	Jarque-Bera (JB):	65.456
Skew:	0.316	Prob(JB):	6.11e-15
Kurtosis:	3.227	Cond. No.	7.33e+03

Solar Potential and Predicted Median Energy Usage



OLS Regression Results						
=====						
Dep. Variable:	usage_med_predicted	R-squared:	0.486			
Model:	OLS	Adj. R-squared:	0.486			
Method:	Least Squares	F-statistic:	659.6			
Date:	Mon, 05 May 2025	Prob (F-statistic):	0.00			
Time:	21:39:31	Log-Likelihood:	-37184.			
No. Observations:	3487	AIC:	7.438e+04			
Df Residuals:	3481	BIC:	7.442e+04			
Df Model:	5					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	1598.3977	649.557	2.461	0.014	324.846	2871.949
bedrooms	1179.2339	260.855	4.521	0.000	667.790	1690.677
bathrooms	1805.2062	434.448	4.155	0.000	953.408	2657.005
price	-0.0063	0.001	-5.266	0.000	-0.009	-0.004
square_feet	4.2112	0.080	52.910	0.000	4.055	4.367
usage_percap_predicted	2.4485	0.316	7.759	0.000	1.830	3.067
=====						
Omnibus:	1738.944	Durbin-Watson:	0.864			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	342489.063			
Skew:	1.249	Prob(JB):	0.00			
Kurtosis:	51.487	Cond. No.	6.08e+05			

Conclusions

This analysis explored residential energy consumption patterns across Los Angeles by integrating multiple data sources into predictive models and analyzing the features derived. The bubble map showed **affluent neighborhoods/cities seems to use more usage per capita**. Further analysis with box charts revealed a **positive relationship between income and per capita energy usage**, particularly at income thresholds of around \$56,000 and \$99,000. Moreover, the scatter plot showed that locations with **greater solar potential exhibited lower energy usage**.

The **final multivariate linear regression model**, which included bedrooms, bathrooms, square footage, price, and predicted per capita usage as predictors, explained approximately 49% of the variance in household energy usage, with all variables statistically significant ($p\text{-value} < 0.001$). **Each additional bathroom increased annual consumption by about 1805 kWh, and more bedrooms and larger square footage consistently correlated with higher energy demand**. Interestingly, **higher home prices slightly reduced energy usage when controlling the home size**. This suggests that higher-priced homes (keeping the size same) likely feature improved insulation, efficient systems, and better infrastructure, thus lowering overall energy usage.

These results highlight that structural characteristic, such as size and layout, are primary drivers of energy consumption, suggesting that energy efficiency policies should focus less on home price and more on physical attributes like square footage and number of rooms.

Maintainability and Extensibility

The **UCLA_data_processing_and_API.ipynb** notebook handles data cleaning and geolocation. It is well-organized, with clearly labeled preprocessing steps and relative file paths, making it straightforward to rerun with updated datasets or file structures. Testing code segments are provided to verify the validity of GeoAPI responses, ZIP-code assignments, and name normalization. If future datasets include additional or shifted columns, the logic can be adjusted. For Google's Geocoding API calls, basically change which keys to get from the JSON. The **Craigslist_Scraping.ipynb** notebook is responsible for housing data collection. It uses modular scraping logic, with isolated components for parsing, range splitting, and retrying. Key operations are clearly separated into function blocks, and test code is provided to verify correctness of individual listings and data batch integrity. It is extensible for additional filters or new regions, and any future adjustments to Craigslist's HTML structure can be implemented with edits to the parsing logic without disrupting the overall framework.

The **Predicting_energy.ipynb** notebook is responsible with model training steps follow consistent preprocessing and training patterns, with easily adjustable consistent code block format. The code is easily extendable for new features or model updates.